

Use-Case Flow Specification – View Bus ETA

System: Public Bus Live Tracking & Crowding Estimator

Use Case ID: UC-02

Use Case Name: View Bus ETA

Primary Actor: Commuter

Supporting Actor: Bus GPS Device/System

Description

The commuter selects a bus and the system displays its estimated arrival time at upcoming stops based on the latest available GPS data.

Preconditions

1. The commuter has access to the bus tracking platform.
2. The selected bus route exists in the system.
3. Recent GPS telemetry is available for the selected bus.
4. The system is operational.

Postconditions

1. The commuter is shown the estimated arrival times for upcoming stops.
2. The displayed ETA is based on the latest available bus location and telemetry.
3. The system does not display an outdated ETA as current information.

Main Success Scenario

1. The commuter opens the Public Bus Live Tracking & Crowding Estimator.
2. The system displays available bus routes.
3. The commuter selects a bus route.
4. The system displays active buses operating on the selected route.
5. The commuter selects a bus.
6. The system retrieves the latest GPS location and speed of the selected bus.
7. The system identifies the upcoming stops on the selected route.

8. The system calculates the estimated arrival time for each upcoming stop.
9. The system displays the bus's current location and ETA for the upcoming stops.
10. The commuter views the estimated arrival information.
11. The use case ends successfully.

Alternate Flow

6a. GPS Data Unavailable

1. The system attempts to retrieve the latest GPS data.
2. The system determines that valid recent GPS data is unavailable.
3. The system displays a message indicating that the live ETA is temporarily unavailable.
4. The system does not display a misleading or outdated ETA.
5. The commuter may select another active bus or try again later.
6. The use case ends.